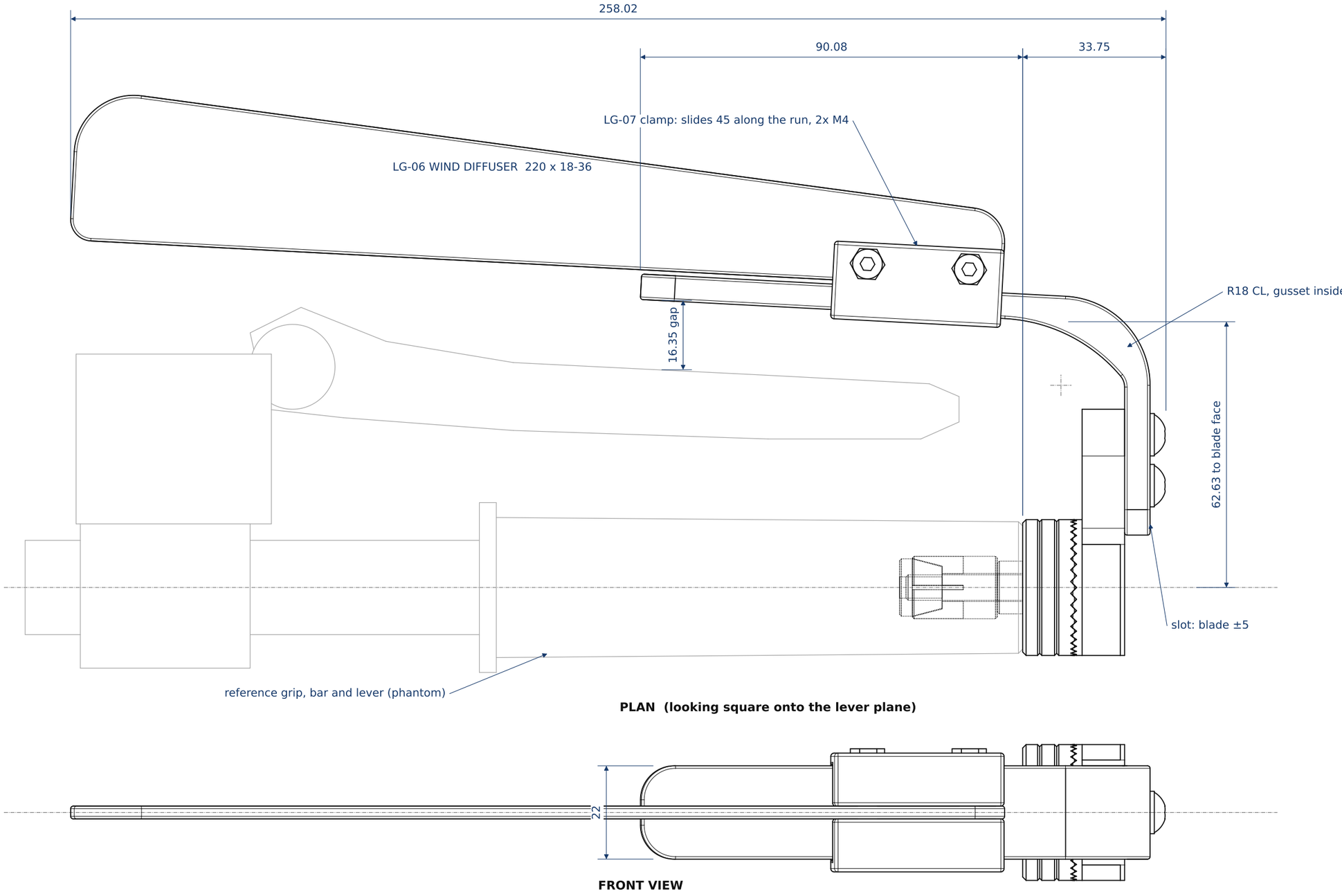
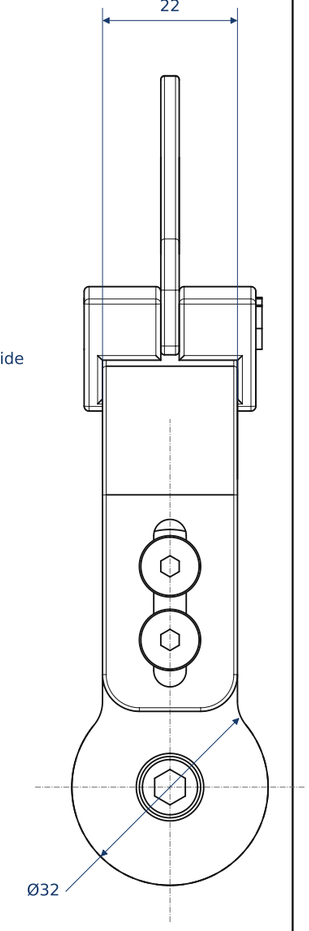


PLAN (looking square onto the lever plane)

FRONT VIEW



BAR-END VIEW

NOTES

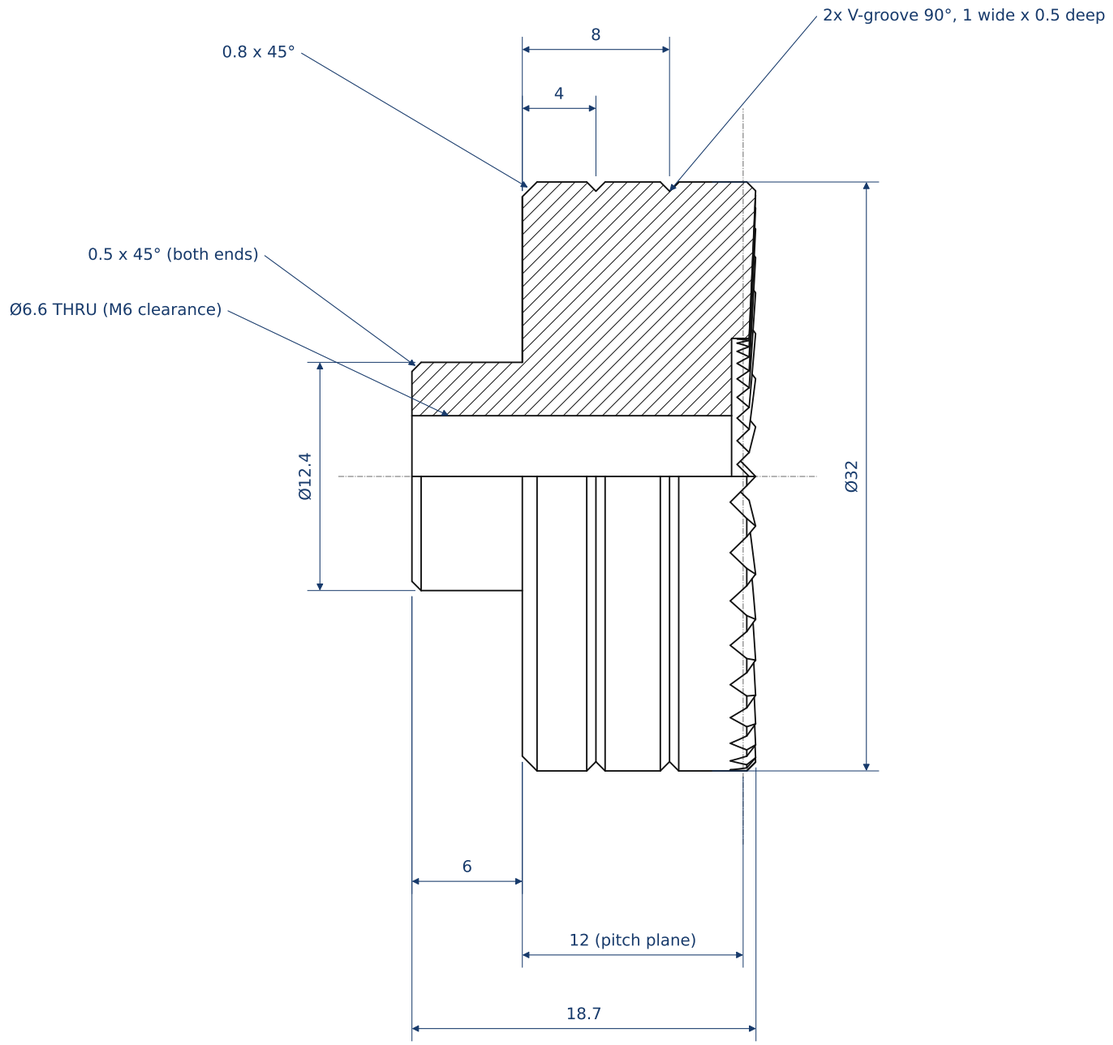
1. Drawing plane = lever plane. Turn the arm on its 36-tooth serration (10° steps) to put the blade on the lever line.
2. Clearances shown for the reference sport-bike lever; keep ≥ 10 mm all round, re-check at full lever travel.
3. Mass per side ≈ 123 g, 189 g with the wind diffuser.
4. Left side: same parts, arm turned over (every part is symmetric about the lever plane).

RACING LEVER GUARD / UNIVERSAL BAR-END FIT

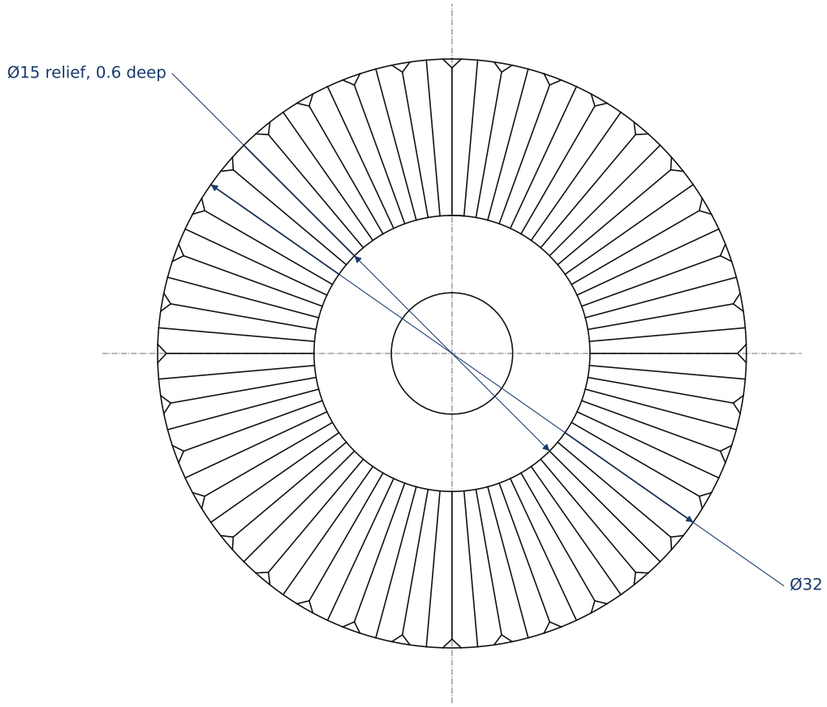
GENERAL ARRANGEMENT

LG-00

MATERIAL				FINISH	
see part sheets				see part sheets	
SCALE	UNITS	QTY / SIDE	GENERAL TOLERANCES	PROJECTION	SHEET
1 : 1	mm	1 set	ISO 2768-mK		1 / 6
SOURCE				DATE	BREAK SHARP EDGES
cad/lever_guard.py (CadQuery), drawings.py				2026-10-23	0.2 - 0.5



HALF SECTION A-A



VIEW ON SERRATION (from outboard)

NOTES

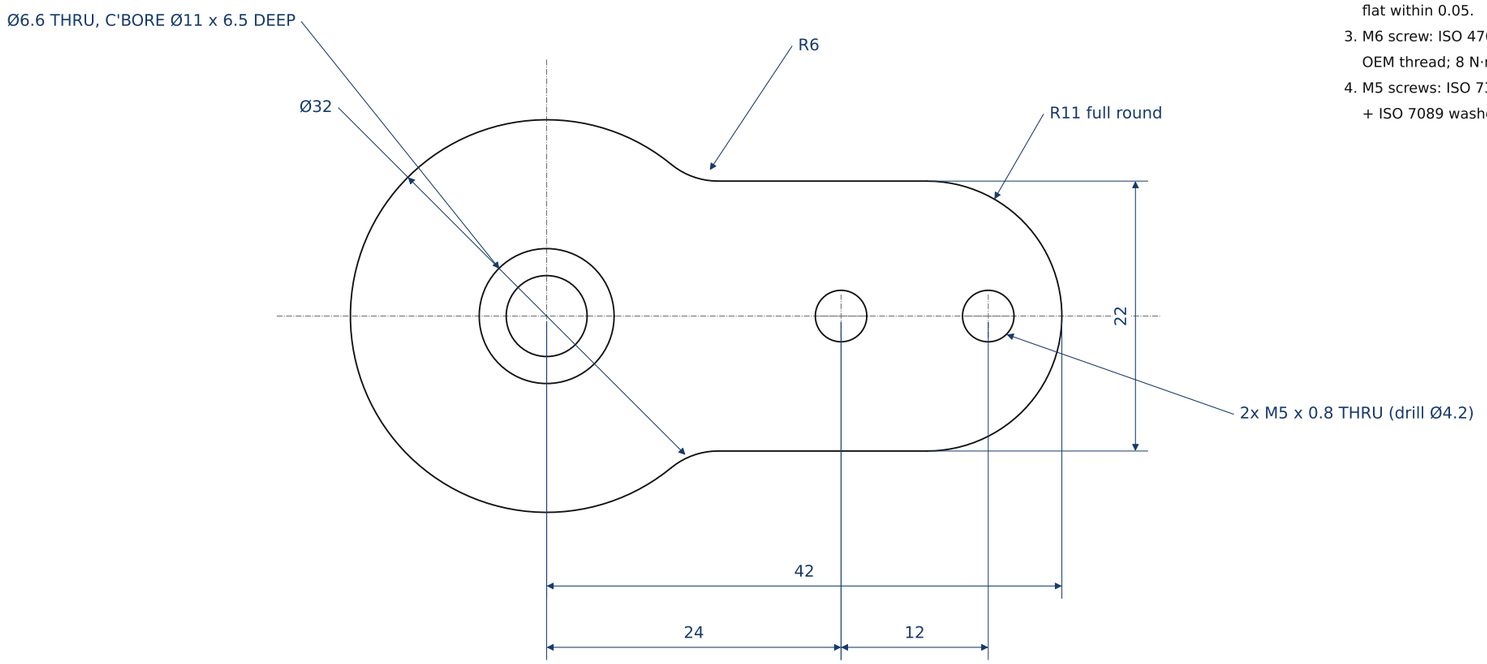
1. Hirth face serration: 36 teeth, 90° included, crests and roots radial through the axis, pitch plane at 12 from the bar-end face, toothed from Ø15 to Ø32: tooth height 1.4 at Ø32. Mates with LG-02 (offset half a tooth).
2. Variant LG-01b: same part without the spigot, for bars with a threaded OEM bar-end insert.
3. Spigot Ø12.4 centres in any bar with ID ≥ 12.5; the expander (LG-04/05) sits behind it.
4. Hard anodise after machining; mask nothing. Serration may be cut with a 90° V-cutter.

RACING LEVER GUARD / UNIVERSAL BAR-END FIT

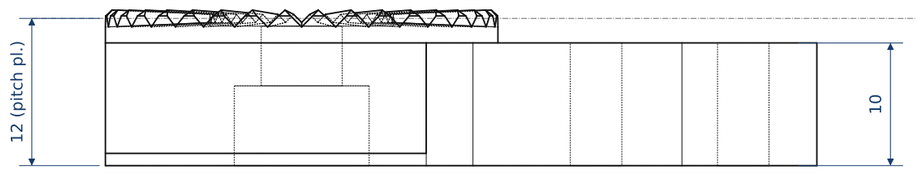
BAR-END ADAPTER

LG-01

MATERIAL			FINISH		
EN AW-6061-T6			Hard anodise, colour to order		
SCALE	UNITS	QTY / SIDE	GENERAL TOLERANCES	PROJECTION	SHEET
3 : 1	mm	1	ISO 2768-mK		2 / 6
SOURCE			DATE		BREAK SHARP EDGES
cad/lever_guard.py (CadQuery), drawings.py			2026-10-23		0.2 - 0.5



OUTBOARD FACE



PLAN (lever plane), serration toward the bar

NOTES

1. Hirth serration as LG-01: 36 teeth, 90°, pitch plane on the hub's inner face. Sits half a tooth out of phase with LG-01 so crests meet roots: 10° indexing.
2. Neck edges R1. Outboard face is the blade seat: flat within 0.05.
3. M6 screw: ISO 4762 M6 x 45 (expander) or to suit the OEM thread; 8 N·m, medium threadlocker.
4. M5 screws: ISO 7380 M5 x 14 (x 16 with LG-03b) + ISO 7089 washers, 4 N·m.

RACING LEVER GUARD / UNIVERSAL BAR-END FIT

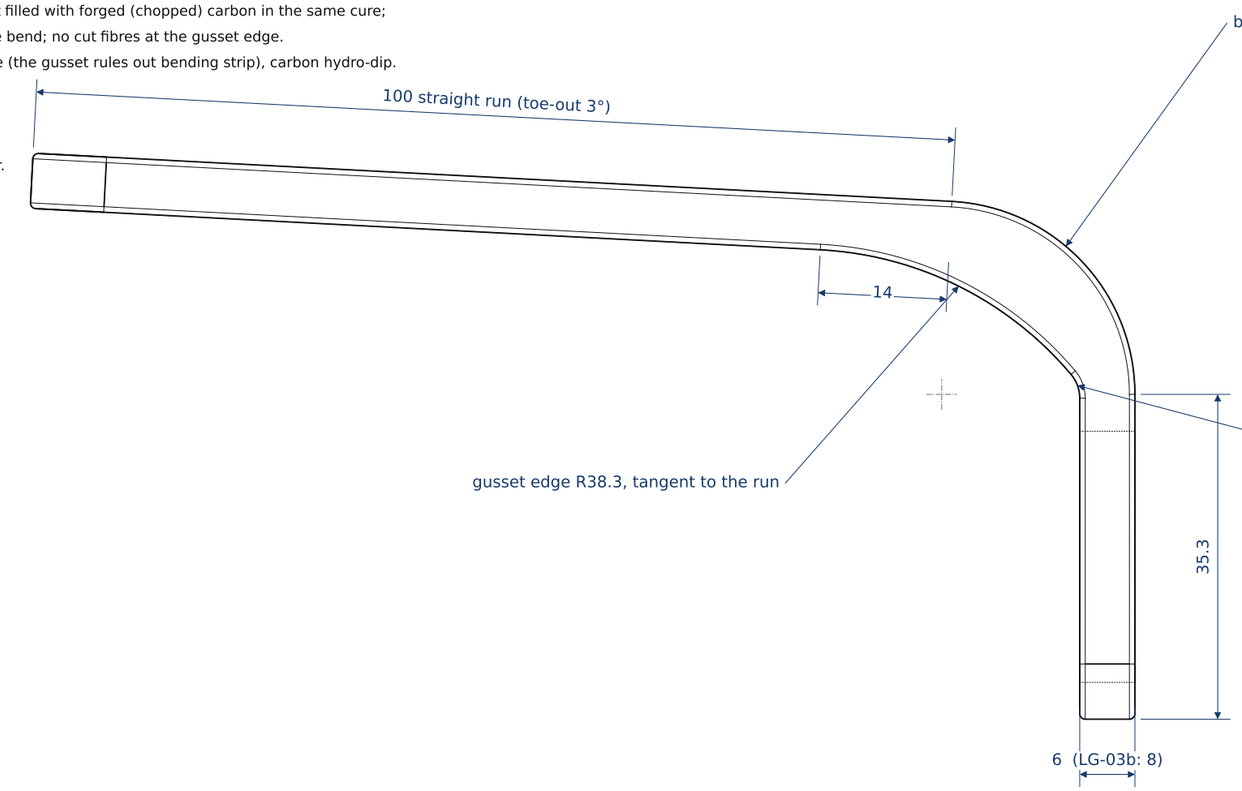
GUARD ARM

LG-02

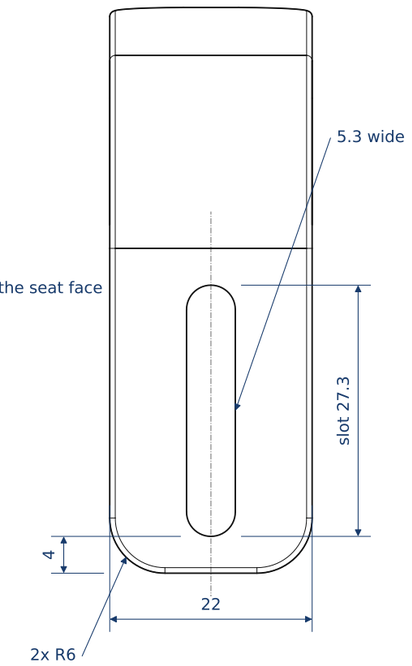
MATERIAL				FINISH	
EN AW-6061-T6				Hard anodise, colour to order	
SCALE	UNITS	QTY / SIDE	GENERAL TOLERANCES	PROJECTION	SHEET
2 : 1	mm	1	ISO 2768-mK		3 / 6
SOURCE				DATE	BREAK SHARP EDGES
cad/lever_guard.py (CadQuery), drawings.py				2026-10-23	0.2 - 0.5

NOTES

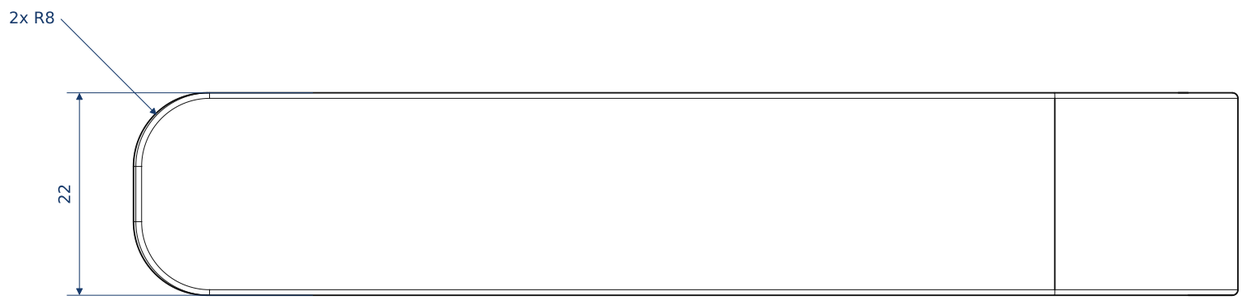
1. Centre-line length 162.6 mm: 35.3 straight + 87° bend on R18 + 100 straight.
2. Gusset inside the bend: section 6 -> 9.7 mm across the bend, about 2.6x the bending strength where a fall loads the blade.
   Its edge is tangent all along (blend into the seat face, arc, run-out on the run): no corner to start a crack.
3. LG-03 carbon: skin 6 ± 0.2 moulded over a male tool, gusset filled with forged (chopped) carbon in the same cure;
   CNC-trim the outline and slot. Lay plies continuous round the bend; no cut fibres at the gusset edge.
4. LG-03 aluminium option: machine from 6061-T6 25 mm plate (the gusset rules out bending strip), carbon hydro-dip.
5. LG-03b 3D print: same outline, 8 thick,
   PA12-CF / PETG-CF, lying flat (lever plane on the bed),
   6 walls, 50 % gyroid; hydro-dip or wrap for the carbon finish.
6. Slot lets the blade slide ±5 mm toward / away from the lever.



PLAN (lever plane)



REAR PLATE (outboard)



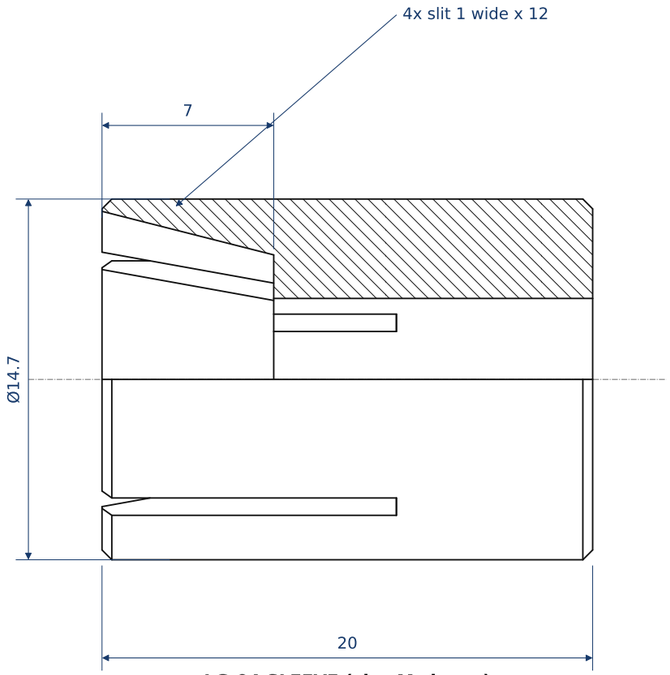
FRONT VIEW (outer face)

RACING LEVER GUARD / UNIVERSAL BAR-END FIT

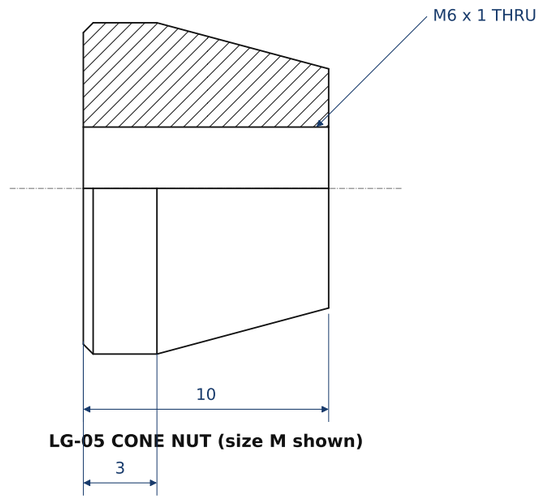
GUARD BLADE WITH BEND GUSSET

LG-03

MATERIAL: 3K 2x2 twill carbon / epoxy, forged-carbon gusset				FINISH: Gloss clear coat, edges R0.6	
SCALE: 3 : 2	UNITS: mm	QTY / SIDE: 1	GENERAL TOLERANCES: ISO 2768-mK	PROJECTION	SHEET: 4 / 6
SOURCE: cad/lever_guard.py (CadQuery), drawings.py				DATE: 2026-10-23	BREAK SHARP EDGES: 0.2 - 0.5



LG-04 SLEEVE (size M shown)



LG-05 CONE NUT (size M shown)

SIZES (all lengths as drawn: sleeve 20, recess 7, cone land 3)

Size	Bar ID	Sleeve OD	Recess Ø face / bottom	Cone Ø big / small	Taper (half)
S	13 - 15	12.7	11.7 / 8.4	11.5 / 8	14°
M	15 - 17	14.7	13.7 / 10.15	13.5 / 9.75	15°
L	17 - 19	16.7	15.7 / 12.15	15.5 / 11.75	15°

NOTES

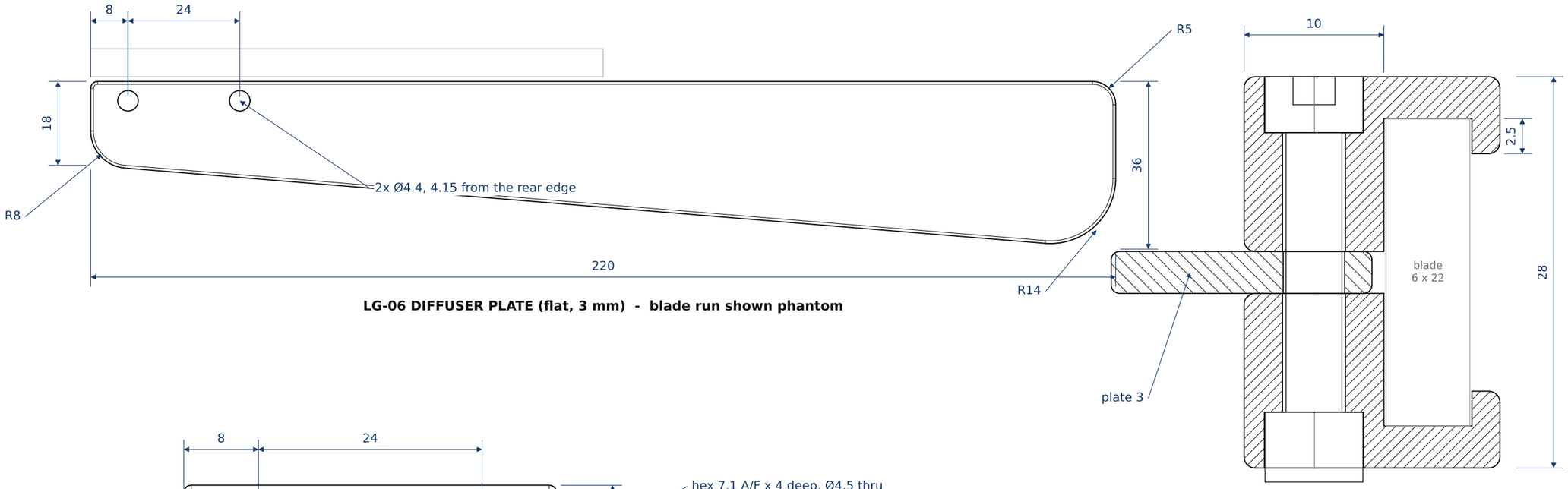
1. Bolt chain: M6 x 45 (ISO 4762) through LG-02 and LG-01 into the cone nut: 8.5 mm thread engagement at fit-up.
2. Tightening pulls the cone into the slit sleeve, which grips the bar bore. 8 N·m. Hold the arm square on its serration while tightening.
3. Pick the size from the bar's measured inside diameter.
4. Sleeve can be printed in TPU 95A (100 % infill) or cut from NBR tube; cone may be steel for higher load.

RACING LEVER GUARD / UNIVERSAL BAR-END FIT

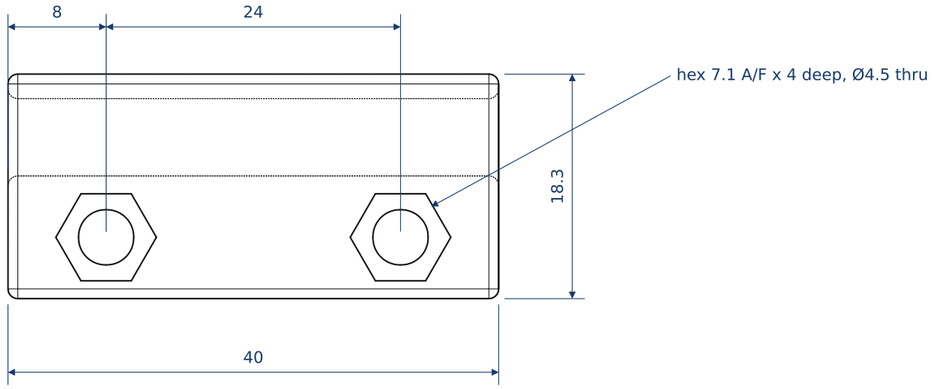
EXPANDER SET

LG-04 / LG-05

MATERIAL: LG-04 TPU 95A or NBR 70A; LG-05 6061-T6				FINISH: LG-05 clear anodise or zinc	
SCALE: 4 : 1	UNITS: mm	QTY / SIDE: 1 sleeve + 1 cone	GENERAL TOLERANCES: ISO 2768-mK	PROJECTION	SHEET: 5 / 6
SOURCE: cad/lever_guard.py (CadQuery), drawings.py				DATE: 2026-10-23	BREAK SHARP EDGES: 0.2 - 0.5



LG-06 DIFFUSER PLATE (flat, 3 mm) - blade run shown phantom



LG-07 CLAMP HALF, outer face (2:1)

SECTION THROUGH A SCREW (3:1)

NOTES

1. LG-07 is one part used twice: the lower half is the upper turned over.
2. Fit with the guard on the bike: halves above and below the blade run, plate strip between their front blocks, 2x M4 x 25 ISO 4762 + DIN 985 nyloc nuts, 2.5 N·m. No holes in the blade.
3. Tightening squeezes the halves onto the blade's top and bottom edges and pinches the plate. Loosen to slide: 15-60 mm from the bend.
4. Plate: 3 mm 3K carbon cut from sheet, or 5052 aluminium + hydro-dip.
5. With the diffuser fitted keep the blade slot at -3 or more: at -5 the clamp lip comes within 8 mm of the reference lever.
6. Check mirror and reservoir clearance before sliding it inboard.

RACING LEVER GUARD / UNIVERSAL BAR-END FIT

WIND DIFFUSER + SLIDING CLAMP

LG-06 / LG-07

MATERIAL				FINISH	
LG-06 3K carbon plate 3 mm; LG-07 EN AW-6061-T6				LG-07 hard anodise, colour to order	
SCALE	UNITS	QTY / SIDE	GENERAL TOLERANCES	PROJECTION	SHEET
1:1 / 2:1 / 3:1	mm	1 + 2 halves	ISO 2768-mK		6 / 6
SOURCE				DATE	BREAK SHARP EDGES
cad/lever_guard.py (CadQuery), drawings.py				2026-10-23	0.2 - 0.5